WITH HIGH PROBABILITY THE RANDOM SUBGRAPH ADMITS NO SMALL TUTTE OBSTRUCTION IN THE WHOLE WINDOW OF EDGE PROBABILITIES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the source pins the publisher class scrartcl, which the delivery wrapper does not accept, so this card uses sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/retained-preamble.tex, which supplies the amsart baseline and then carries the author's own newtheorem and macro declarations (pinned source lines 30-57) verbatim. No macro was invented and no mathematical symbol was changed.

Editorial note. Bibliography provenance: the source carries no thebibliography environment and ends with bibliographystyle abbrv plus bibliography perc (author source lines 1691-1692), so the bibliography is resolved with BibTeX from the identical copy staged with the same arXiv version under sources/product-graph-matchings-2404.14020v4/perc.bib; no entry was edited and the bibliography style is the author's own abbrv.

Editorial note. Two labels the counted proof invokes, lem:three_edges and lem:small-boundary-sets, and the displayed range for p are navigation pointers only; the content they name is declared as named standard prerequisites and is not reproduced here. The same is true of the Gallai-Edmonds relation eq:Gallai-Edmonds, which enters only through the decomposition the surrounding subsection sets up.

Editorial note. The counted statement is delivered together with the Tutte-violation display and the definition of u_0 as context spans, both quoted verbatim from the author's own text.

Allowed standard prerequisites (editorial).

- The Tutte-violation relation and the threshold u_0. Stated by the author immediately before the two lemmas (source lines 656-668), delivered verbatim as context spans of this card. Fixes the property that the obstruction set must satisfy and the size threshold below which the lemma excludes it.
- The minimality lemma for obstructions and the counting lemmas for sets with a small edge boundary. The author's lemmas at source lines 703-705 and 714-716, whose proofs are delivered by this lane's cards for those statements. Supplies the two ingredients whose combination gives the size bound for the obstruction.

Source reference. Original lemma 6.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 703–703. Navigation pointer only: the counted proof uses this minimality lemma in its first step; its own statement and proof are delivered by this lane's separate card for that statement and are not part of this card.

Source reference. Original lemma 6.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 714–714. Navigation pointer only: the counted proof uses this counting lemma in its second step; its own statement and proof are delivered by this lane's separate card and are not part of this card.

Source reference. Original equation 4.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the range of p displayed there is what makes the estimate uniform over the whole window; the display is not part of this card.

Source reference. Original equation 3.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 266–266. Navigation pointer only: the counted proof consumes the Gallai-Edmonds decomposition that the surrounding subsection sets up; the relation itself is not part of this card.

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Source reference. Original lemma 6.5: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 772–772. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 6.6: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 842–842. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Lemma 0.1. *Whp, simultaneously for every $p \in [p_-, p_+]$ with $\delta(G_p) \geq 1$, the graph G_p has no set U with $1 \leq |U| \leq u_0$ satisfying (1).*

$$\operatorname{odd}(G_p - U) \geq |U| + 2 \tag{1}$$

$$u_0 \coloneqq \frac{n}{d^{M^5/p_-}}.$$

Proof of Theorem 0.1. Recall that whp every G_p under consideration is connected and that G has even order. In particular, the empty set does not satisfy (1). Choose a set U satisfying (1) of minimum cardinality, put $u = |U|$, and let $K_1, \dots, K_m$ be the odd components of $G_p - U$. By Original lemma 6.2 every $v \in U$ has neighbours in at least three of these components. Every K_i which is an isolated vertex in $G_p - U$ has a G_p-neighbour in U, because $\delta(G_p) \geq 1$.

Suppose that $u \leq u_0$. Whp, at most one K_i has order larger than u_1 by Original lemma 6.5. Let B be the union of the remaining odd components. Note that there are at least $u + 1$ of them and every $v \in U$ has neighbours in at least two components of B. If $|B| < u_1$, then U and B satisfy the assumptions of Original lemma 6.6, a contradiction. If $|B| \geq u_1$, then a greedy union of its components has size between u_1 and $2u_1 < n/2$. In this case U together with this union of components satisfy the assumptions of Original lemma 6.5, a contradiction. □